

United International University

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Experiment No. 01

Name of the Experiment: Determination of the refractive index of a liquid by plane mirror and pin method using a convex lens.

Theory:

If a convex lens is placed on a few drops of liquid on a plane mirror, then on squeezing the liquid into the space between the mirror and the lens, a Plano-concave liquid lens is formed. The curved surface of this liquid lens has the same radius of curvature as the surface of the lens with which it is in contact. Thus we have a combination of two lenses – one of glass and the other of liquid, which behaves as a convergent lens. If f_2 be the focal length of the combination, then we have the relation

$$\frac{1}{f_2} = \frac{1}{f_1} + \frac{1}{F} \dots \dots \dots (1)$$

where f_1 and F are the focal lengths of the convex lens and the liquid lens respectively.

Correcting for the sign of F which is negative, we get

$$\frac{1}{f_2} = \frac{1}{f_1} - \frac{1}{F}$$

$$\frac{1}{F} = \frac{1}{f_1} - \frac{1}{f_2} \dots \dots \dots (2)$$

Determining f_2 and f_1 experimentally, we can calculate F from relation (2).

The focal length of the plano-concave liquid lens is also given by relation

$$\frac{1}{F} = (\mu - 1) \left(\frac{1}{r} - \frac{1}{r'} \right) = (\mu - 1) \frac{1}{r}$$

($r' = \infty$, the lower face of the liquid lens being a plane)

According to sign convention, both F and r are negative. Thus,

$$\mu = 1 + \frac{r}{F} \dots \dots \dots (3)$$

Where μ is the refractive index of the liquid.

Finding r , the radius of curvature of the lower surface of the convex lens i.e. the surface in contact with the liquid, and knowing F from relation (2), the refractive index of the liquid, μ can be found out by using relation (3)

Note: The radius of curvature of the surface of the lens is given by,

$$r = \frac{a^2}{6h} + \frac{h}{2}$$

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Apparatus:

- A convex lens
- A plane mirror
- Pin/pointer with its tip painted
- Spherometer
- Slide calipers
- Stand
- Some experimental liquid (water or glycerin)

Experimental Data:

(A) Measurement of h

Reading on	No. of obs.	LSR, x (cm.)	CSD	LC (cm.)	Value of CSR, $y=LC \times CSD$ (cm.)	Total reading, $x+y$ (cm.)	Mean (cm.)
Base plate	1	0	1	0.001	0.001	0.001	0.06
	2	0	86		0.086	0.086	
	3	0	93		0.093	0.093	
Lens surface	1	0.2	2		0.002	0.202	0.211
	2	0.2	6		0.006	0.206	
	3	0.2	25		0.025	0.225	

$h = \text{Reading on lens surface} - \text{Reading on the base plate} = 0.161 \text{ cm}$

Measurement of 'a': Mean value of $a = \frac{a_1 + a_2 + a_3}{3} = \frac{3.8 + 3.9 + 3.9}{3} = 3.867 \text{ cm}$

Therefore, radius of curvature of the spherical surface, $r = \frac{a^2}{6h} + \frac{h}{2} = 16.581 \text{ cm}$

(B) Table for the thickness of convex lens t

No. of obs.	MSR, x (cm.)	VSD	Vernier Constant, VC (cm.)	Value of VSR, $y=VC \times VSD$ (cm.)	Total reading, $x+y$ (cm.)	Mean thickness (cm.)	Instrumental Error	Correct thickness t (cm.)
1	0.8	6	0.005	0.03 0.03	0.83	0.83	0	0.83
2	0.8	4		0.02	0.82			
3	0.8	8		0.04	0.84			

(C) Determination of the focal lengths

No. of obs.	Distance between the pin and the face of the lens (without the liquid), h_1 (cm.)	Focal length of the convex lens, $f_1 = h_1 + \frac{t}{3}$ (cm.)	Mean f_1 (cm.)	Distance between the pin and the face of the lens (with the liquid), h_2 (cm.)	Focal length of the combination, $f_2 = h_2 + \frac{t}{3}$ (cm.)	Mean f_2 (cm.)	Focal length of the liquid lens, $F = \frac{f_2 f_1}{f_2 - f_1}$ (cm.)
1	0.2	0.476	9.942	16.3	16.576	16.542	24.963
2	9.6	9.876		16.4	16.676		
3	16.2	10.476		16.1	16.376		

Calculation:

$$\therefore \mu = 1 + \frac{r}{F} = 1.664$$

Result:

The refractive index of the given liquid is, $\mu = 1.664$.

Discussions:

Q: What is refractive index? Write down the refractive index of water and compare (% difference and % accuracy) your result.

The refractive index (n) of a medium is defined as the ratio of the speed of light in vacuum to the speed of light in that medium. $n = \frac{c}{v}$

Standard $n_w = 1.33$

$$\text{Diff} = \left| \frac{1.64 - 1.33}{1.33} \right| \times 100$$

$$= \cancel{25.90\%} 25.11\%$$

$$\text{Accuracy} = (1 - \text{diff.})$$

$$= 1 - 25.11\%$$

$$= 74.89\%$$

$$\approx 75\%$$

Q: What is the physical significance of refractive index?

Refractive index physically represents how much a medium slows down light and how strongly it bends light compared to vacuum.

Physics Laboratory

Q: What is the speed of light through the liquid whose refractive index you have determined?

$$n = 1.664 \quad c = 3 \times 10^8$$

$$\text{So, } n = \frac{c}{v} \Rightarrow v = \frac{c}{n} = \frac{3 \times 10^8}{1.664} = 18.02 \times 10^7 \text{ m/s}$$

Q: What is radius of curvature? Is it possible to find the refractive index of the given liquid without using a spherometer?

The radius of curvature of a curved surface is the radius of the imaginary sphere of which the surface forms a part. Yes it is possible. Refractive index of a liquid can be determined without a spherometer using methods like apparent depth, critical angle method or refractometer.

Q: Calculate the Least Count of the given spherometer.

$$\text{Pitch of screw} = 1 \text{ mm}$$

$$\text{Number of divisions on circular scale} = 100$$

$$\therefore \text{least count} = \frac{1}{100} = 0.01 \text{ mm}$$